Aneurysm CTA Dataset

A simple guide to what is in each folder and file

Folder	~/Desktop/Personal Projects/Aneurysm_dataset
Checked on	1 November 2026
Total size	about 26 GB

1. The dataset in one paragraph

This dataset has **3D CT scans of the head** and **masks** that mark brain aneurysms. An aneurysm is a small bulge in a blood vessel wall. It can burst (rupture) and cause bleeding in the brain. The scans are **CT angiography (CTA)**: a dye is injected so blood vessels look bright. Each mask is a 3D map of the same size as its scan: 1 = aneurysm, 0 = not aneurysm. The usual task is **segmentation**: teach a model to draw the mask from the scan.

2. Simple words used in this document

Word	Meaning
CT / CTA	CT = X-ray based 3D scan. CTA = CT with dye so vessels are bright.
Scan	The CT image. Stored as a 3D grid of numbers.
Mask	Same size grid as the scan. 1 = aneurysm, 0 = everything else.
Voxel	One 3D pixel. Like a tiny box.
Slice	One flat 2D picture from the 3D scan.
Spacing	Real size of one voxel, in mm (e.g. 0.45 x 0.45 x 0.4 mm).
HU	Hounsfield Unit. CT brightness scale. Air = -1000, water = 0, bone = +700 and above.
NIfTI (.nii.gz)	Standard file type for 3D medical images. '.gz' means compressed.
Orientation	Which way the axes point. LPS = Left, Posterior (back), Superior (up).
Ruptured	The aneurysm has burst.
IA	Intracranial aneurysm (aneurysm inside the head).

3. Folder map

Name	What it is	Count	Size
internal_Tr01/	CT scans (images)	99 scans	13 GB
internal_Tr01.zip	Same 99 scans, zipped (duplicate)	1 file	13.7 GB
internal_annotation/	Masks for internal cases	1,338 masks	344 MB
internal_annotation.7z	Same masks, compressed	1 file	0.9 MB
external_annotation/	Masks for external test cases (2 other hospitals)	138 masks	52 MB
external_annotation.7z	Same masks, compressed	1 file	84 KB
external_instances.csv	Patient info for the external cases	138 rows	6 KB
aneurysm_eda.ipynb	Our EDA notebook	-	-
scan_and_mask_visualisation.ipynb	Our simple viewing notebook (12 views)	-	-
eda_cache/mask_stats.csv	Saved statistics for all 1,338 internal masks	-	-

What you have vs. what you do not have

Group	Masks	Scans	Usable together
Internal train (Tr0001-Tr1186)	1,186	99 (Tr0001-Tr0099)	99
Internal test (Ts0001-Ts0152)	152	0	0
External A (ExtA0001-ExtA0071)	71	0	0
External B (ExtB0001-ExtB0067)	67	0	0

You chose to download only the first scan part (Tr01). So **99 scan + mask pairs** are ready to use. All other masks have no scan in the folder.

4. File type and how to open it

Every scan and mask is a **.nii.gz** file. The file name links a scan to its mask:

```
internal_Tr01/Tr0004.nii.gz        <- scan
internal_annotation/Tr0004.nii.gz  <- its mask (same name)
```

Open with Python (nibabel):

```
import nibabel as nib, numpy as np
nii = nib.load("internal_Tr01/Tr0004.nii.gz")
img = np.asanyarray(nii.dataobj)          # 3D array of HU values
spacing = nii.header.get_zooms()          # voxel size in mm
```

Tip: use `np.asanyarray(nii.dataobj)`, not `get_fdata()`. get_fdata() makes a float64 copy (about 1.9 GB for a big scan).

5. Scans: internal_Tr01/

Item	Value
Number of scans	99 (Tr0001 to Tr0099)
Scan type	CT angiography of the head (some also include the neck)
Shape	512 x 512 x N (N = 265 to 942 slices)
Voxel spacing	In-plane 0.37 to 0.64 mm. Between slices: mostly 0.4 mm (also 0.5, 0.625, 0.7)
Values	HU, from -1024 to 3071
Data type	int16 (Tr0096 is int32)
Orientation	LPS for all
File size	about 75 to 260 MB each (compressed)
Memory when loaded	about 140 to 500 MB each (int16)

Typical HU values in these scans

Thing	HU
Air	about -1000
Brain	about 20 to 45
Vessels with dye	about 200 to 500
Aneurysm (inside)	about 240 to 260 on average
Bone	700 and above

6. Internal masks: internal_annotation/

Item	Value
Number of masks	1,338 = 1,186 train (Tr) + 152 test (Ts)
Values	0 = background, 1 = aneurysm (only one label)
Shape	Same as its scan (512 x 512 x N). N = 176 to 1,310 over all masks
Data type	uint16 (empty test masks are int16)
Orientation	LPS (one exception: Tr0370 is ALS)
Match with scan	All 99 pairs checked: same shape, same spacing, same position

Train vs. test

	Train (Tr)	Test (Ts)
Cases	1,186	152
With aneurysm	1,186 (all)	102
Without aneurysm (empty mask)	0	50
1 aneurysm	1,041	85
2 aneurysms	118	13
3 or more	27	4
Median aneurysm volume	74 mm^3	106 mm^3 (cases with aneurysm)
Median size (as a ball)	5.2 mm across	5.9 mm across
Most aneurysms (5% to 95%)	2.8 to 11.4 mm across	-

Aneurysms are **very small**. In a scan, only about 1 voxel in 100,000 is aneurysm. Note: multiple aneurysms share the same label (1). To count them, split the mask into separate blobs.

7. External masks: external_annotation/

A second test set from **two other hospitals**. It is used to check if a model still works on scans from hospitals it never saw in training. Only masks are here, no scans.

Item	Set A (ExtA)	Set B (ExtB)
Hospital (institution_id)	7	8
Number of masks	71	67
Slice size	varies: 365 x 365 to 870 x 870 (both sets)	
Number of slices	368 to 1,314 (both sets)	
Voxel spacing	0.488 x 0.488 x 0.45 mm	0.529 x 0.529 x 0.45 mm
Orientation	LPS	LPS
Data type	int16 or uint16 (mixed)	int16 or uint16 (mixed)

Different from internal: slice size is not always 512 x 512, and spacing is the same within each set. These were probably resampled and cropped before release.

8. Patient info: external_instances.csv

One row per external patient. The internal data has **no** file like this.

Column	Meaning	Example
subset	Which external set	external_test_A
instance_id	File name (without .nii.gz)	ExtA0001
institution_id	Hospital number	7
age	Patient age in years	48
gender	male / female	male
is_ruptured	Has the aneurysm burst? yes / no	yes
num_IAs	Number of aneurysms (0 = none)	1

Summary of the CSV

	Set A	Set B
Patients	71	67
No aneurysm (num_IAs = 0)	24	22
1 aneurysm	44	40
2 or more aneurysms	3	5
Ruptured (yes)	29	25
Age (min / median / max)	14 / 56 / 86	26 / 62 / 88
Female / male	39 / 32	34 / 33

Every CSV row has a mask file, and every mask file has a CSV row. Not yet checked: whether each mask has the same number of aneurysms as num_IAs says.

9. How a scan and its mask fit together

```
Scan  = the photo           (512 x 512 x N numbers, HU)
Mask  = tracing on top      (512 x 512 x N numbers, 0 or 1)
Same position in both = same point in the head
```

Because they line up voxel by voxel, you can lay the mask over the scan directly. No alignment step is needed.

10. Known issues

Issue	Where	What to do
Only 99 scans present	internal_Tr01	Use 99 pairs; download more parts if needed
No aneurysm-free scans in your 99	Tr0001-Tr0099	Model never sees a normal scan; sample background patches
Different orientation	Tr0370 mask (ALS)	Reorient all files to one orientation when loading
Different data type	Tr0096 scan (int32, 0.625 mm slices)	Convert to int16 / resample
Tiny specks in masks (35 blobs under 20 voxels)	e.g. Tr0734 (21 blobs), Tr0749, Tr0320, Tr0837	Remove blobs under ~20 voxels
Spacing is not the same for all scans	All	Resample to one spacing (e.g. 0.4 to 0.5 mm)
Duplicate data (13.7 GB)	internal_Tr01.zip	Can be deleted after checking the folder
External masks have no scans	external_annotation	Download external scans to use them

11. Tips for working with this data

- **Memory:** a 900-slice scan uses about 0.5 GB. Load one case at a time.
- **Viewing:** show HU from -150 to 550 to see vessels well.
- **Small target:** aneurysms are tiny, so zoom in or crop around them.
- **Training:** use patches around the aneurysm plus random background patches.
- **Splitting:** split by case (patient), never by slice.
- **Orientation:** use `nib.as_closest_canonical()` so all cases face the same way.

12. Notebooks in this folder

File	What it does
scan_and_mask_visualisation.ipynb	Easy notebook: 12 ways to view one scan and its mask (slice, overlay, contour, zoom, MIP, slider, 3D, histogram).
aneurysm_eda.ipynb	Full EDA: inventory, scan details, mask statistics, intensity, sample views, quality checks.
eda_cache/mask_stats.csv	Saved per-mask numbers (size, blobs, spacing) so the EDA notebook runs fast.